



Universidad Tecnológica Equinoccial

Escuela de Tecnologías

Carrera de Desarrollo de Software

Programación III

Tema: Despliegue Nestjs - Proyecto Integrador “Consultorios Médicos”

Trabajo N-002

Fecha: 12/01/2025

Presentado por:

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Desplegar aplicación backend Nestjs en azure virtual machine

Registrar todos los pasos pantallas, scripts y código empleado en un archivo pdf y adjuntar a la tarea

1.- Objetivo

Realizar el despliegue completo de una aplicación NestJS en una máquina virtual de Azure (Ubuntu 20.04 o superior), aplicando buenas prácticas de DevOps, incluyendo un despliegue manual inicial y la automatización mediante CI/CD con GitHub Actions.

Descripción de la actividad

Cada grupo deberá realizar las siguientes actividades:

1. Crear una máquina virtual en Azure con sistema operativo Ubuntu.

projecto-integrador-app

Máquina virtual

Ayuda para copiar esta máquina virtual en cualquier región

Administrar esta VM con CLI de Azure

Ayuda para copiar esta máquina virtual en cualquier región

Conectar

Iniciar

Reiniciar

Detener

Hibernar

Captura

Eliminar

Actualizar

Abrir en dispositivos móviles

Comentarios

CLI / PS

Información esencial

Grupo de recursos (mover) : rg-proyecto-integrador

Estado : En ejecución

Ubicación : West US 3 (Zona 1)

Suscripción (mover) : Azure for Students

Id. de suscripción : 989a962d-5a12-4855-9d9f-793269f3e291

Zona de disponibilidad : 1

Sistema operativo : Linux (ubuntu 22.04)

Tamaño : Standard B2ats v2 (2 vcpu, 1 GiB de memoria)

IP pública de NIC principal : 20.163.4.48

1 direcciones IP públicas asociadas

Red virtual/subred : proyecto-integrador-app-vnet/default

Nombre DNS : Sin configurar

Estado de mantenimiento : -

Hora de creación : 12/1/2026, 2:11 a.m. UTC

Etiquetas (editar) : Agregar etiquetas

projecto-integrador-app | Configuración de red

¿Cuáles son los requisitos para conectar o desasociar una interfaz de red?

Enumere todas mis interfaces de red para [0].

+1

Buscar

Información general

Registro de actividad

Control de acceso (IAM)

Etiquetas

Diagnosticar y solucionar problemas

Visualizador de recursos

Conectar

Redes

Configuración de red

Equilibrio de carga

Grupos de seguridad de la aplicación

Administrador de red

Configuración

Disponibilidad y escala

Seguridad

Copia de seguridad y recuperación

Agregue o quite favoritos presionando Ctrl+Shift+F

Grupo de seguridad de red proyecto-integrador-app-nsg (conectado a networkInterface: proyecto-integrador-app10_z1)

Afecta a 0 subredes, 1 interfaces de red

+ Crear ACL del puerto

Buscar reglas

Origen == todo

Destino == todo

Protocolo == todo

Acción == todo

Puerto == todo

Prioridad	Nombre	Puerto	Protocolo	Origen	Destino	Acción
Reglas de puerto de entrada (6)						
300	SSH	22	TCP	Cualquiera	Cualquiera	Allow
310	allow-http	8080	TCP	Cualquiera	Cualquiera	Allow
330	http	443	TCP	Cualquiera	Cualquiera	Allow
65000	AllowVnetInBound	Cualquiera	Cualquiera	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Cualquiera	Cualquiera	AzureLoadBalancer	Cualquiera	Allow
65500	DenyAllInBound	Cualquiera	Cualquiera	Cualquiera	Cualquiera	Deny
Reglas de puerto de salida (4)						
320	exit-http	80	TCP	Cualquiera	Cualquiera	Allow
65000	AllowVnetOutBound	Cualquiera	Cualquiera	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Cualquiera	Cualquiera	Cualquiera	Internet	Allow
65500	DenyAllOutBound	Cualquiera	Cualquiera	Cualquiera	Cualquiera	Deny

2. Conectarse por SSH a la máquina virtual y preparar el entorno instalando Node.js, npm, Git y PM2.

- Conexión a la máquina virtual por medio de ssh

```
karling23@proyecto-integrad x + v
The authenticity of host '20.163.4.48 (20.163.4.48)' can't be established.
ED25519 key fingerprint is SHA256:vk604D3w0qpeCP803HweWTRLF0dYSTeg*7PK1WHzKls.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.163.4.48' (ED25519) to the list of known hosts.
karling23@20.163.4.48's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1044-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Jan 12 02:17:18 UTC 2026

System load:  0.47      Processes:    128
Usage of /:   6.3% of 28.89GB Users logged in: 0
Memory usage: 38%      IPv4 address for eth0: 10.1.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

17 updates can be applied immediately.
16 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

karling23@proyecto-integrador-app:~$ |
```

- Instalación de Node.js, npm, git y PM2

```
karling23@proyecto-integrador-app:~$ sudo apt update && sudo apt upgrade -y
sudo apt install -y nodejs npm git
sudo npm install -g pm2
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
17 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  dirmngr gnupg gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm gpgv libglib2.0-0 libglib2.0-bin libglib2.0-data
17 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
16 standard LTS security updates
Need to get 36.4 MB of archives.
After this operation, 549 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gpg-wks-client amd64 2.2.27-3ubuntu2.5 [62.7 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 dirmngr amd64 2.2.27-3ubuntu2.5 [293 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gpg-wks-server amd64 2.2.27-3ubuntu2.5 [57.6 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gnupg-utils amd64 2.2.27-3ubuntu2.5 [309 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gnupg-agent amd64 2.2.27-3ubuntu2.5 [209 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gpg amd64 2.2.27-3ubuntu2.5 [519 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gpgconf amd64 2.2.27-3ubuntu2.5 [94.3 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gnupg-l10n all 2.2.27-3ubuntu2.5 [54.5 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gnupg all 2.2.27-3ubuntu2.5 [315 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gpgsm amd64 2.2.27-3ubuntu2.5 [197 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gpgv amd64 2.2.27-3ubuntu2.5 [137 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libglib2.0-data all 2.72.4-0ubuntu2.7 [5178 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libglib2.0-bin amd64 2.72.4-0ubuntu2.7 [80.9 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libglib2.0-0 amd64 2.72.4-0ubuntu2.7 [1467 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsodium23 amd64 1.0.18-1ubuntu0.22.04.1 [164 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libxslt1.1 amd64 1.1.34-4ubuntu0.22.04.5 [165 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 snapd amd64 2.73+ubuntu22.04 [32.3 MB]
```

- Instalación de postgres

```
karling23@proyecto-integrador-app:~$ sudo apt install -y postgresql postgresql-contrib
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcommon-sense-perl libjson-perl libjson-xs-perl libllvm14 libpq5 libtypes-serialiser-perl postgresql-14 postgresql-client-14 postgresql-client-common
  postgresql-common ssl-cert
Suggested packages:
  postgresql-doc postgresql-doc-14
The following NEW packages will be installed:
  libcommon-sense-perl libjson-perl libjson-xs-perl libllvm14 libpq5 libtypes-serialiser-perl postgresql postgresql-14 postgresql-client-14
  postgresql-client-common postgresql-common postgresql-contrib ssl-cert
0 upgraded, 13 newly installed, 0 to remove and 0 not upgraded.
Need to get 42.0 MB of archives.
After this operation, 160 MB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 libcommon-sense-perl amd64 3.75-2build1 [21.1 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 libjson-perl all 4.04000-1 [81.8 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 libtypes-serialiser-perl all 1.01-1 [11.6 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libjson-xs-perl amd64 4.040-0ubuntu0.22.04.1 [87.0 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libllvm14 amd64 1:14.0.0-1ubuntu1.1 [24.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpq5 amd64 14.20-0ubuntu0.22.04.1 [153 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 postgresql-client-common all 238 [29.6 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 postgresql-client-14 amd64 14.20-0ubuntu0.22.04.1 [1250 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 ssl-cert all 1.1.2 [17.4 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 postgresql-common all 238 [169 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 postgresql-14 amd64 14.20-0ubuntu0.22.04.1 [16.2 MB]
Get:12 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 postgresql all 14+238 [3288 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 postgresql-contrib all 14+238 [3292 B]
Fetched 42.0 MB in 1s (62.2 MB/s)
Preconfiguring packages ...
Selecting previously unselected package libcommon-sense-perl:amd64.
(Reading database ... 91941 files and directories currently installed.)
Preparing to unpack .../00-libcommon-sense-perl_3.75-2build1_amd64.deb ...
Unpacking libcommon-sense-perl:amd64 (3.75-2build1) ...
Selecting previously unselected package libjson-perl.
Preparing to unpack .../01-libjson-perl_4.04000-1_all.deb ...
Unpacking libjson-perl (4.04000-1) ...
Selecting previously unselected package libtypes-serialiser-perl.
Preparing to unpack .../02-libtypes-serialiser-perl_1.01-1_all.deb ...
Unpacking libtypes-serialiser-perl (1.01-1) ...
```

- Configuración de postgres

```
postgres=# \c proyectoint_db_pg
You are now connected to database "proyectoint_db_pg" as user "postgres".
proyectoint_db_pg=# CREATE USER proyectoint_user_pg WITH PASSWORD 'proyectoint_pass_123';
ALTER ROLE proyectoint_user_pg SET client_encoding TO 'utf8';
ALTER ROLE proyectoint_user_pg SET default_transaction_isolation TO 'read committed';
ALTER ROLE proyectoint_user_pg SET timezone TO 'UTC';

GRANT ALL PRIVILEGES ON DATABASE proyectoint_db_pg TO proyectoint_user_pg;
GRANT ALL ON SCHEMA public TO proyectoint_user_pg;

ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT ALL ON TABLES TO proyectoint_user_pg;

ALTER DATABASE proyectoint_db_pg OWNER TO proyectoint_user_pg;
ALTER SCHEMA public OWNER TO proyectoint_user_pg;

GRANT CREATE ON SCHEMA public TO proyectoint_user_pg;
CREATE ROLE
ALTER ROLE
ALTER ROLE
ALTER ROLE
GRANT
GRANT
ALTER DEFAULT PRIVILEGES
ALTER DATABASE
ALTER SCHEMA
GRANT
proyectoint_db_pg=# |
```

- Instalación de mongoDB.

```
karling23@proyecto-integrador-app:~$ wget -qO - https://www.mongodb.org/static/pgp/server-6.0.asc | sudo apt-key add -
Warning: apt-key is deprecated. Manage keyring files in trusted.gpg.d instead (see apt-key(8)).
OK
karling23@proyecto-integrador-app:~$ echo "deb [ arch=amd64,arm64 ] https://repo.mongodb.org/apt/ubuntu focal/mongodb-org/6.0 multiverse" | sudo tee /etc/ap
t/sources.list.d/mongodb-org-6.0.list
deb [ arch=amd64,arm64 ] https://repo.mongodb.org/apt/ubuntu focal/mongodb-org/6.0 multiverse
karling23@proyecto-integrador-app:~$ echo "deb http://security.ubuntu.com/ubuntu focal-security main" | sudo tee /etc/apt/sources.list.d/focal-security.list
deb http://security.ubuntu.com/ubuntu focal-security main
karling23@proyecto-integrador-app:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease
Get:5 https://repo.mongodb.org/apt/ubuntu focal/mongodb-org/6.0 InRelease [4009 B]
Get:6 http://security.ubuntu.com/ubuntu focal-security InRelease [129 kB]
Get:7 https://repo.mongodb.org/apt/ubuntu focal/mongodb-org/6.0/multiverse arm64 Packages [113 kB]
Get:8 https://repo.mongodb.org/apt/ubuntu focal/mongodb-org/6.0/multiverse amd64 Packages [117 kB]
Get:9 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages [3564 kB]
Get:10 http://security.ubuntu.com/ubuntu focal-security/main Translation-en [518 kB]
Get:11 http://security.ubuntu.com/ubuntu focal-security/main amd64 c-n-f Metadata [14.4 kB]
Fetched 4458 kB in 2s (1791 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
W: https://repo.mongodb.org/apt/ubuntu/dists/focal/mongodb-org/6.0/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DE
PRECATION section in apt-key(8) for details.
karling23@proyecto-integrador-app:~$ sudo apt install -y libssl1.1
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  libssl1.1
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 1323 kB of archives.
After this operation, 4131 kB of additional disk space will be used.
Get:1 http://security.ubuntu.com/ubuntu focal-security/main amd64 libssl1.1 amd64 1.1.1f-1ubuntu2.24 [1323 kB]
Fetched 1323 kB in 1s (1020 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libssl1.1:amd64.
(Reading database ... 93833 files and directories currently installed.)
```

```
karling23@proyecto-integrador-app:~$ sudo systemctl start mongod
karling23@proyecto-integrador-app:~$ sudo systemctl enable mongod
Created symlink /etc/systemd/system/multi-user.target.wants/mongod.service → /lib/systemd/system/mongod.service.
karling23@proyecto-integrador-app:~$ sudo systemctl status mongod
● mongod.service - MongoDB Database Server
   Loaded: loaded (/lib/systemd/system/mongod.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-01-12 02:35:53 UTC; 13s ago
     Docs: https://docs.mongodb.org/manual
   Main PID: 11651 (mongod)
    Memory: 139.5M
      CPU: 275ms
    CGroup: /system.slice/mongod.service
            └─11651 /usr/bin/mongod --config /etc/mongod.conf

Jan 12 02:35:53 proyecto-integrador-app systemd[1]: Started MongoDB Database Server.
Jan 12 02:35:53 proyecto-integrador-app mongod[11651]: {"t":{"$date":"2026-01-12T02:35:53.961Z"},"s":"I", "c":"CONTROL", "id":7484500, "ctx":"","msg":"E
```

- Configuración de mongoDB

```
karling23@proyecto-integrador-app:~$ mongo --directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.10
Connecting to:
Using MongoDB: 6.0.27
Using Mongosh: 2.5.10

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

test> use admin
switched to db admin
admin> db.createUser({
...   user: "proyectooint_user_mongo",
...   pwd: "proyectooint_user_mongo_123",
...   roles: [{ role: "root", db: "admin" }]
... })
{ ok: 1 }
admin> use proyectooint_db_mongo
switched to db proyectooint_db_mongo
proyectooint_db_mongo>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
proyectooint_db_mongo>
karling23@proyecto-integrador-app:~$ sudo systemctl restart mongod
karling23@proyecto-integrador-app:~$ sudo systemctl start mongod
sudo systemctl enable mongod
sudo systemctl status mongod
● mongod.service - MongoDB Database Server
   Loaded: loaded (/lib/systemd/system/mongod.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-01-12 02:42:17 UTC; 11s ago
     Docs: https://docs.mongodb.org/manual
   Main PID: 11837 (mongod)
    Memory: 209.5M
      CPU: 502ms
    CGroup: /system.slice/mongod.service
            └─11837 /usr/bin/mongod --config /etc/mongod.conf

Jan 12 02:42:17 proyecto-integrador-app systemd[1]: Started MongoDB Database Server.
Jan 12 02:42:17 proyecto-integrador-app mongod[11837]: {"t":{"$date":"2026-01-12T02:42:17.767Z"},"s":"I", "c":"CONTROL", "id":7484500, "ctx":"","msg":"E
lines 1-12/12 (END)
```

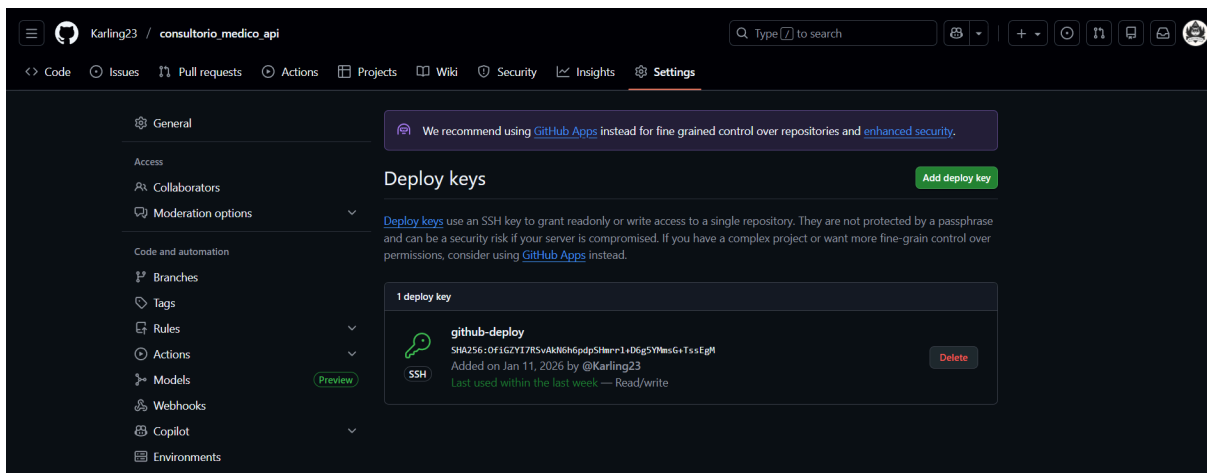
3. Clonar un proyecto NestJS desde GitHub utilizando claves SSH (Deploy Keys).

- Creación de clave pública para clonar repositorio

```
usage: sudo -E [-AADMNS] [-F file] [-C type] [-C num] [-D directory] [-g group] [-H host] [-p prompt] [-W directory] [-T timeout] [-u user] file ...
karling23@proyecto-integrador-app:~$ sudo su
root@proyecto-integrador-app:/home/karling23# cd
root@proyecto-integrador-app:~# cd /opt
bash: cd: /opt: No such file or directory
root@proyecto-integrador-app:~# cd /opt
root@proyecto-integrador-app:/opt# sudo fallocate -l 4G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile

echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
Setting up swapspace version 1, size = 4 GiB (4294963200 bytes)
no label, UUID=9efc2e7b-8c5f-475a-8441-4dbee5f13f26
/swapfile none swap sw 0 0
root@proyecto-integrador-app:/opt# ssh-keygen -t ed25519 -C "github-deploy" -f ~/.ssh/github_deploy
Generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/github_deploy
Your public key has been saved in /root/.ssh/github_deploy.pub
The key fingerprint is:
SHA256:0fiCZV17RSvAkN6h6pdpSHmr1+D6gSYMmsG+TssEgM github-deploy
The key's randomart image is:
+--[ED25519 256]--+
..
.o .
. = .
|E+ o o o .
|B o o = S
|*O o + * .
|**= O . o
|**oO o .
|OB*
+----[SHA256]-----
root@proyecto-integrador-app:/opt# cat ~/.ssh/github_deploy.pub
ssh-ed25519 AAAAC3NzaC1lZD11NTE5AAAAIC870iLV39jyMYyr94N3VpYo5Bz0pRHuERR0oh3MtW8D github-deploy
root@proyecto-integrador-app:/opt# GIT_SSH_COMMAND='ssh -i ~/.ssh/github_deploy' git clone git@github.com:Karling23/consultorio_medico_api.git
Cloning into 'consultorio_medico_api'...
The authenticity of host 'github.com (140.82.113.4)' can't be established.
```

- Configuración de clave en repositorio



- Clonación de repositorio en la máquina virtual

```
root@proyecto-integrador-app:/opt# GIT_SSH_COMMAND='ssh -i ~/.ssh/github_deploy' git clone git@github.com:Karling23/consultorio_medico_api.git
Cloning into 'consultorio_medico_api'...
The authenticity of host 'github.com (140.82.113.4)' can't be established.
ED25519 key fingerprint is SHA256:0fiCZV17RSvAkN6h6pdpSHmr1+D6gSYMmsG+TssEgM.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Enter passphrase for key '/root/.ssh/github_deploy':
remote: Enumerating objects: 482, done.
remote: Counting objects: 100% (31/31), done.
remote: Compressing objects: 100% (30/30), done.
remote: Total 482 (delta 2), reused 4 (delta 1), pack-reused 451 (from 1)
Receiving objects: 100% (482/482), 419.02 KiB | 2.48 MiB/s, done.
Resolving deltas: 100% (300/300), done.
root@proyecto-integrador-app:/opt# cd consultorio_medico_api/
```

4. Configurar las variables de entorno mediante un archivo .env (no versionado).

- Configuración archivo .env

```
root@proyecto-integrador-app:/opt/consultorio_medico_api# nano .env
root@proyecto-integrador-app:/opt/consultorio_medico_api# nano .env
```

```
GNU nano 6.2 .env
DB_HOST=localhost
DB_PORT=5432
DB_USER=proyectoint_user_pg
DB_PASS=proyectoint_pass_123
DB_NAME=proyectoint_db_pg

JWT_SECRET=supersecret
JWT_EXPIRES_IN=3600s

MONGO_URI=mongodb://proyectoint_user_mongo:proyectoint_user_mongo_123@localhost:27017/proyectoint_db_mongo

MAIL_USER=clalquinga@gmail.com
MAIL_PASS=qsbr uqor ewby zuup

SENDGRID_API_KEY=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
SENDGRID_SENDER=clalquinga@gmail.com
```

5. Construir y ejecutar la aplicación NestJS utilizando PM2.

- Instalación y construcción del proyecto

```
root@proyecto-integrador-app:/opt/consultorio_medico_api# npm install
added 1 package, and audited 803 packages in 2s

161 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
root@proyecto-integrador-app:/opt/consultorio_medico_api# npm run build

> consultorio-medico_api@0.0.1 build
> nest build
```

- Funcionamiento de aplicación exitosa

```
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/recetas/:id, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/recetas/:id, PATCH} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/recetas/:id, DELETE} route +1ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RoutesResolver] PacientesController {/pacientes}: +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/pacientes, POST} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/pacientes, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/pacientes/:id, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/pacientes/:id, PUT} route +1ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/pacientes/:id, DELETE} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RoutesResolver] ConsultoriosController {/consultorios}: +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/consultorios, POST} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/consultorios, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/consultorios/:id, GET} route +1ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/consultorios/:id, PUT} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/consultorios/:id, DELETE} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RoutesResolver] CitasMedicasController {/citas-medicas}: +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/citas-medicas, POST} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/citas-medicas, GET} route +1ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/citas-medicas/:id, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/citas-medicas/:id, PUT} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/citas-medicas/:id, DELETE} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RoutesResolver] HistorialClinicoController {/historial-clinico}: +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/historial-clinico, POST} route +1ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/historial-clinico, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/historial-clinico/:id, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/historial-clinico/:id, PUT} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/historial-clinico/:id, DELETE} route +1ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RoutesResolver] MedicamentosController {/medicamentos}: +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/medicamentos, POST} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/medicamentos, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/medicamentos/:id, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/medicamentos/:id, PUT} route +1ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/medicamentos/:id, DELETE} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RoutesResolver] RecetaDetalleController {/receta-detalle}: +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/receta-detalle, POST} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/receta-detalle, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/receta-detalle/:id, GET} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/receta-detalle/:id, PUT} route +1ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [RouterExplorer] Mapped {/receta-detalle/:id, DELETE} route +0ms
[Nest] 13185 - 01/12/2026, 3:09:55 AM LOG [NestApplication] Nest application successfully started +5ms
```


- Inicio de la app con pm2

```
[PM2] Spawning PM2 daemon with pm2_home=/root/.pm2
[PM2] PM2 Successfully daemonized
[PM2] Starting /opt/consultorio_medico_api/dist/main.js in fork_mode (1 instance)
[PM2] Done.
```

id	name	namespace	version	mode	pid	uptime	o	status	cpu	mem	user	watching
0	consultorio-medico-api	default	0.0.1	fork	13251	0s	0	online	0%	33.8mb	root	disabled

```
[PM2] Saving current process list...
[PM2] Successfully saved in /root/.pm2/dump.pm2
[PM2] Init System found: systemd
Platform systemd
Template
[Unit]
Description=PM2 process manager
Documentation=https://pm2.keymetrics.io/
After=network.target

[Service]
Type=forking
User=root
LimitNOFILE=infinity
LimitNPROC=infinity
LimitCORE=infinity
Environment=PATH=/root/.nvm/versions/node/v22.11/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/b
in:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
Environment=PM2_HOME=/root/.pm2
PIDFile=/root/.pm2/pm2.pid
Restart=on-failure

ExecStart=/usr/local/lib/node_modules/pm2/bin/pm2 resurrect
ExecReload=/usr/local/lib/node_modules/pm2/bin/pm2 reload all
ExecStop=/usr/local/lib/node_modules/pm2/bin/pm2 kill

[Install]
WantedBy=multi-user.target

Target path
/etc/systemd/system/pm2-root.service
```

```
Command list
[ 'systemctl enable pm2-root' ]
[PM2] Writing init configuration in /etc/systemd/system/pm2-root.service
[PM2] Making script booting at startup...
[PM2] [-] Executing: systemctl enable pm2-root...
Created symlink /etc/systemd/system/multi-user.target.wants/pm2-root.service → /etc/systemd/system/pm2-root.service.
[PM2] [v] Command successfully executed.
+-----+
[PM2] Freeze a process list on reboot via:
$ pm2 save
[PM2] Remove init script via:
$ pm2 unstartup systemd
```

6. Configurar NGINX como proxy inverso para exponer la API mediante HTTP.

- Configuración de NGINX

```
GNU nano 6.2 /etc/nginx/sites-available/nest-app
server {
    server_name alquinga-consultorio-medico-api.desarrollo-software.xyz;

    location / {
        proxy_pass http://localhost:3000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
    }

    listen 443 ssl; # managed by Certbot
    ssl_certificate /etc/letsencrypt/live/alquinga-consultorio-medico-api.desarrollo-software.xyz/fullchain.pem; # managed by Certbot
    ssl_certificate_key /etc/letsencrypt/live/alquinga-consultorio-medico-api.desarrollo-software.xyz/privkey.pem; # managed by Certbot
    include /etc/letsencrypt/options-ssl-nginx.conf; # managed by Certbot
    ssl_dhparam /etc/letsencrypt/ssl-dhparams.pem; # managed by Certbot
}

server {
    if ($host = alquinga-consultorio-medico-api.desarrollo-software.xyz) {
        return 301 https://$host$request_uri;
    } # managed by Certbot

    listen 80;
    server_name alquinga-consultorio-medico-api.desarrollo-software.xyz;
    return 404; # managed by Certbot
}
```


7. (Opcional) Configurar HTTPS usando Let 's Encrypt.

- Configuración de HTTPS

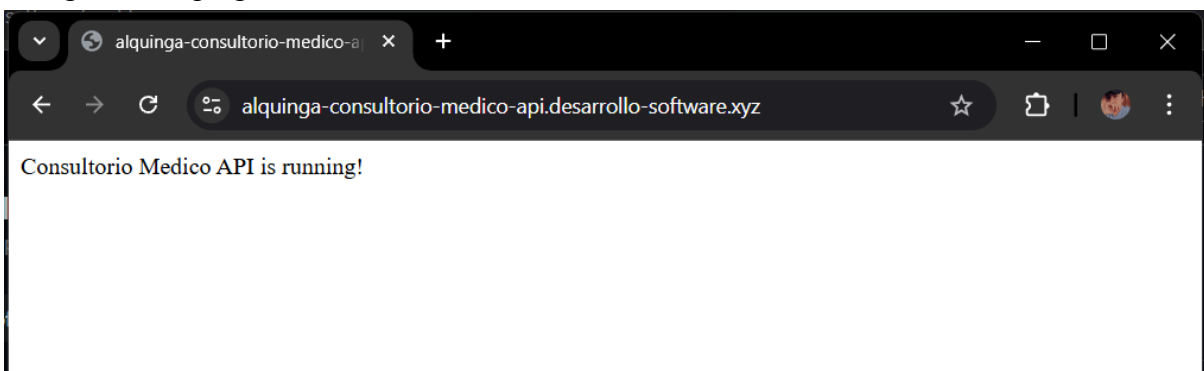
```
root@proyecto-integrador-app:/opt/consultorio_medico_api# sudo certbot --nginx -d alquinga-consultorio-medico-api.desarrollo-software.xyz
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for alquinga-consultorio-medico-api.desarrollo-software.xyz

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/alquinga-consultorio-medico-api.desarrollo-software.xyz/fullchain.pem
Key is saved at: /etc/letsencrypt/live/alquinga-consultorio-medico-api.desarrollo-software.xyz/privkey.pem
This certificate expires on 2026-04-12.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for alquinga-consultorio-medico-api.desarrollo-software.xyz to /etc/nginx/sites-enabled/nest-app
Congratulations! You have successfully enabled HTTPS on https://alquinga-consultorio-medico-api.desarrollo-software.xyz

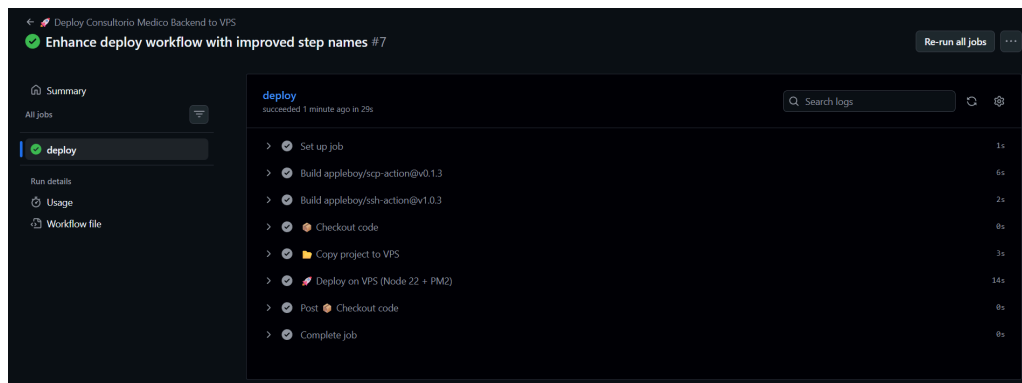
-----
If you like Certbot, please consider supporting our work by:
* Donating to ISRG / Let's Encrypt: https://letsencrypt.org/donate
* Donating to EFF: https://eff.org/donate-le
-----
root@proyecto-integrador-app:/opt/consultorio_medico_api# sudo nano /etc/nginx/sites-available/nest-app
root@proyecto-integrador-app:/opt/consultorio_medico_api#
```

- Programa desplegado correctamente



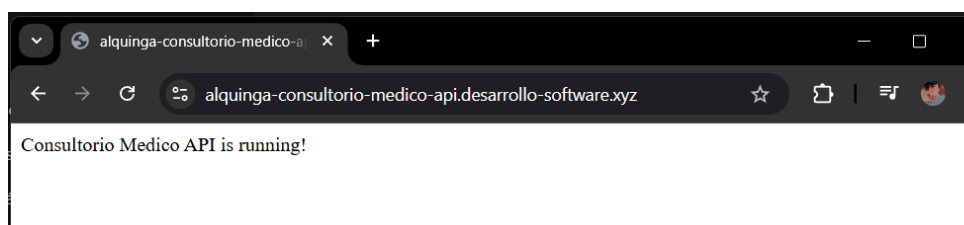
8. Implementar un pipeline de CI/CD con GitHub Actions que permita:

- Creación de pipeline en github

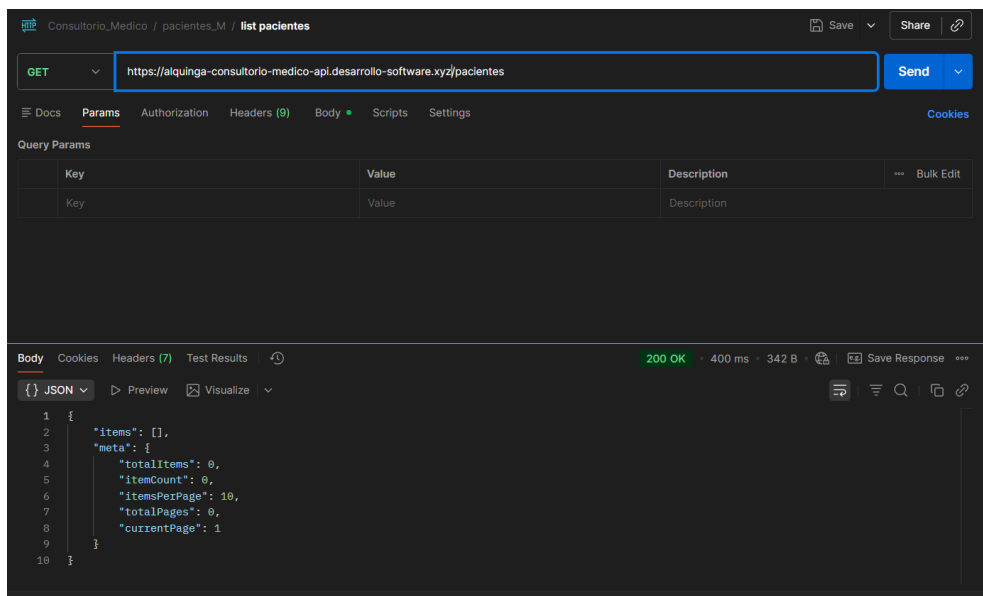


9. Verificar el correcto funcionamiento accediendo a la API desde el navegador o Postman.

- Verificación de funcionamiento web:



postman:



Entregables

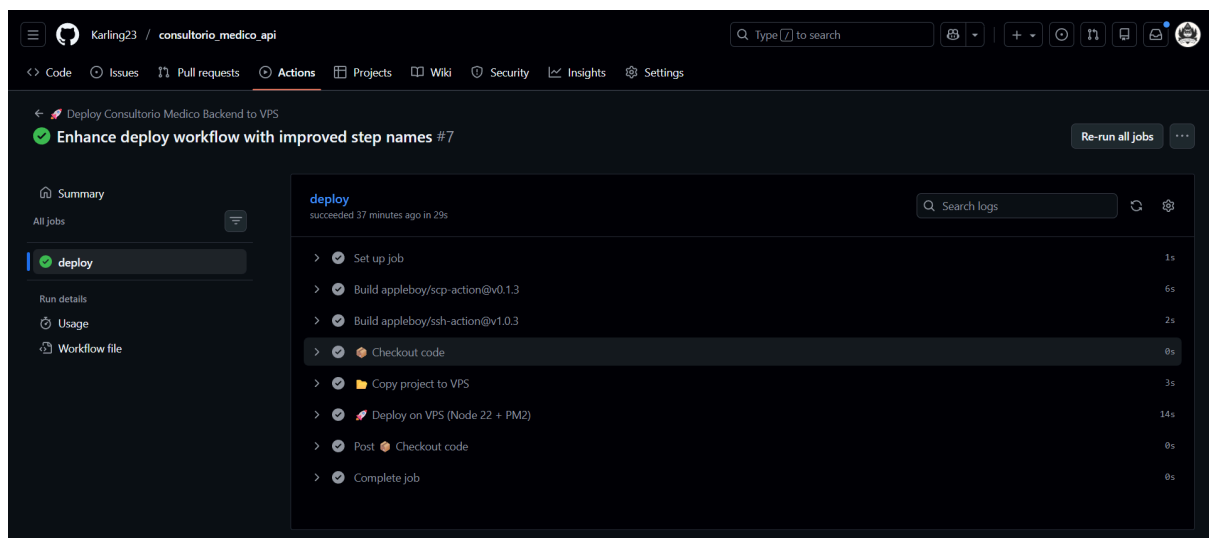
- URL pública de la API desplegada en Azure.

<https://alquinga-consultorio-medico-api.desarrollo-software.xyz/>

- Repositorio GitHub con el workflow de GitHub Actions configurado.

https://github.com/Karling23/consultorio_medico_api/actions

- Captura de pantalla del pipeline ejecutado correctamente.



- Captura de pantalla de PM2 mostrando la aplicación en ejecución.

```
root@proyecto-integrador-app:/opt/consultorio_medico_api# pm2 list
```

id	name	namespace	version	mode	pid	uptime	u	status	cpu	mem	user	watching
0	consultorio-medico-api	default	0.0.1	fork	41992	2h	0	online	0%	121.3mb	root	disabled